

Distribution and Industrial Networks

MPA-MEZ 26/27Z

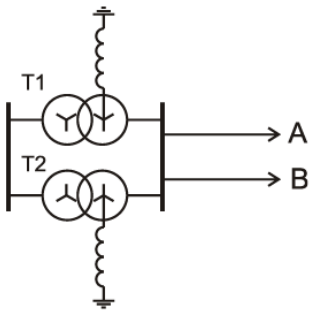
Assignment Solution

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1. Assignment and input data

Cable network 22 kV consists of 2 feeders A and B supplied from a common bus by two HV/MV transformers T1 and T2. Parameters of transformers and lines are

The parameters of the individual feeders are given below. For the calculation, use the parameters corresponding to the given student assignment according to the table in the appendix. Substitute the individual line sections with a Π-element in the calculation. Neglect the effect of the impedance of the 110 kV power system. Neglect the transformer admittance.



System diagram

Calculate:

1. Parameters of the arc suppression coils for compensation of the earth fault current in the network. Consider the active losses of the ASCs as 1.5% of their rated power.
2. The fault current in case of an EF at the specified location, if you consider a leakage current in the network of G = 0.6 µS/km.
3. The zero-sequence voltage and current components that would be measured at feeders A and B by the feeder protection relays during the state of compensation. Plot the calculated values in a phasor diagram.

Quantity	Value
Voltage and frequency	22 kV; 50 Hz
Each transformer	40 MVA; $u_k = 18\%$; $\Delta P_k = 195$ kW
Feeder A	K10; 35 km; fault at 21 km
Feeder B	K2; 10 km
Leakage conductance	0.6 μS/km per phase
ASC losses	1.5% of rated power

Model: $R_f = 0$, unloaded ends, complete Π sections A1 (21 km), A2 (14 km) and B (10 km). HV impedance and transformer admittances are neglected.

The lecture-model assumptions are

$$\overline{Z}_T^{(0)} = \overline{Z}_T^{(1)} = \overline{Z}_T^{(2)}, \qquad \overline{z}'^{(0)} = \overline{z}'^{(1)} = \overline{z}'^{(2)}, \qquad C'_0 = C'_1 = C'_2 = C'.$$

All phasors are RMS values, initially referred to $\overline{E} = U_n/\sqrt{3}\angle 0°$.

2. Arc-suppression coil design

2.1 Total capacitance and uncompensated earth-fault current

Capacitances are per phase.

$$\omega = 2\pi f = 314.1593\,\mathrm{rad/s}, \tag{1}$$

$$E = \frac{U_n}{\sqrt{3}} = 12\,701.706\,\mathrm{V}, \tag{2}$$

$$C_A = 35(0.400) = 14.000\,\mu\mathrm{F}, \tag{3}$$

$$C_B = 10(0.190) = 1.900\,\mu\mathrm{F}, \tag{4}$$

$$C_\Sigma = C_A + C_B = 15.900\,\mu\mathrm{F}, \tag{5}$$

$$I_C = 3\omega C_\Sigma E = \sqrt{3}\omega C_\Sigma U_n = 190.340\,\mathrm{A}. \tag{6}$$

2.2 Required power and selection from the assigned series

For two identical coils:

$$Q_{L,\Sigma} = EI_C = \omega C_\Sigma U_n^2 = 2.417644\,\mathrm{MVAr}, \tag{7}$$

$$Q_{L,\mathrm{each}} = \frac{Q_{L,\Sigma}}{2} = 1.208822\,\mathrm{MVAr}, \tag{8}$$

$$Q_{n,1} = Q_{n,2} = 1.700\,\mathrm{MVAr}. \tag{9}$$

Selected rating: the next higher value in the specified series.

$$I_n = \frac{Q_n}{E} = 133.840\,\mathrm{A}. \tag{10}$$

2.3 Nominal resonance setting and active-loss resistance

$$X_{L,\mathrm{nom}} = \frac{E^2}{Q_{L,\mathrm{each}}} = \frac{U_n^2}{3Q_{L,\mathrm{each}}} = 133.4633\,\Omega, \tag{11}$$

$$L_{\mathrm{nom}} = \frac{X_{L,\mathrm{nom}}}{\omega} = 0.424827\,\mathrm{H}, \tag{12}$$

$$\Delta P_{\mathrm{ASC}} = 0.015Q_n = 25.500\,\mathrm{kW} \quad \text{(each)}, \tag{13}$$

$$R_{\mathrm{ASC}} = \frac{E^2}{\Delta P_{\mathrm{ASC}}} = \frac{U_n^2}{3\Delta P_{\mathrm{ASC}}} = 6326.7974\,\Omega, \tag{14}$$

$$\overline{Z}_{\mathrm{ASC,nom}} = \left(\frac{1}{R_{\mathrm{ASC}}} + \frac{1}{\mathrm{j}X_{L,\mathrm{nom}}}\right)^{-1} = (2.81414 + \mathrm{j}133.40390)\,\Omega. \tag{15}$$

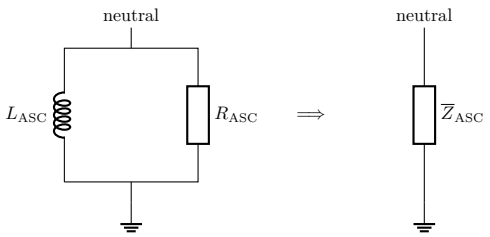


Figure 2. ASC equivalent circuit; zero-sequence impedance: $3\overline{Z}_{\mathrm{ASC}}$.

3. Transformer and cable parameters

3.1 Transformer impedance, referred to 22 kV

$$R_T = \frac{\Delta P_k U_n^2}{S_T^2} = 0.0589875\,\Omega, \tag{16}$$

$$|Z_T| = \frac{u_k}{100}\frac{U_n^2}{S_T} = 2.178000\,\Omega, \tag{17}$$

$$X_T = \sqrt{|Z_T|^2 - R_T^2} = 2.1772011\,\Omega, \tag{18}$$

$$\overline{Z}_T = (0.0589875 + \mathrm{j}2.1772011)\,\Omega, \tag{19}$$

$$\overline{Y}_T = 0. \tag{20}$$

For the two identical transformers in parallel,

$$\overline{Z}_s^{(1)} = \overline{Z}_s^{(2)} = \frac{\overline{Z}_T}{2} = (0.0294938 + \mathrm{j}1.0886005)\,\Omega. \tag{21}$$

$$\overline{Z}_s^{(0)} = \frac{\overline{Z}_T + 3\overline{Z}_{\mathrm{ASC}}}{2}. \tag{22}$$

3.2 Parameters of each complete Π-element

$$\overline{Z}_k = l_k(R'_k + \mathrm{j}\omega L'_k). \tag{23}$$

$$\overline{Y}_k = l_k(g' + \mathrm{j}\omega C'_k), \qquad k \in \{A1, A2, B\}. \tag{24}$$

Cable	R' [Ω/km]	L' [mH/km]	C' [μF/km]	g' [μS/km]
K10 (A)	0.061	0.300	0.400	0.600
K2 (B)	0.443	0.420	0.190	0.600

Section	l [km]	$\overline{Z}_k$ [Ω]	$\overline{Y}_k$ [mS]
A1: S to F	21	$1.281000 + \mathrm{j}1.979203$	$0.012600 + \mathrm{j}2.638938$
A2: F to A end	14	$0.854000 + \mathrm{j}1.319469$	$0.008400 + \mathrm{j}1.759292$
B: S to B end	10	$4.430000 + \mathrm{j}1.319469$	$0.006000 + \mathrm{j}0.596903$

Each endpoint shunt is $\overline{Y}_k/2$. For example, $\overline{Y}_{A1}/2 = (0.006300 + \mathrm{j}1.319469)\,\mathrm{mS}$.

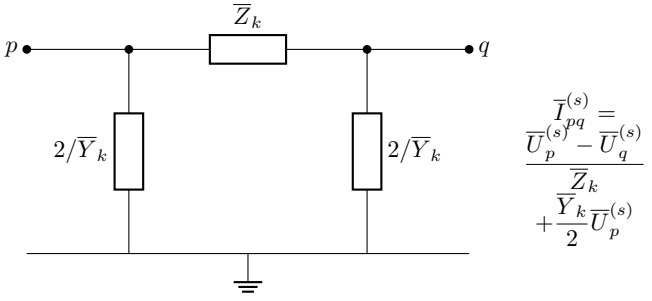


Figure 3. Section Π-element with shunt impedances $2/\overline{Y}_k$.

The section values apply to all three sequence networks.

4. Sequence circuits and successive reductions

4.1 Assemble the three sequence networks

$\overline{E}^{(1)} = \overline{E}$. For sequences 0 and 2, the source is short-circuited.

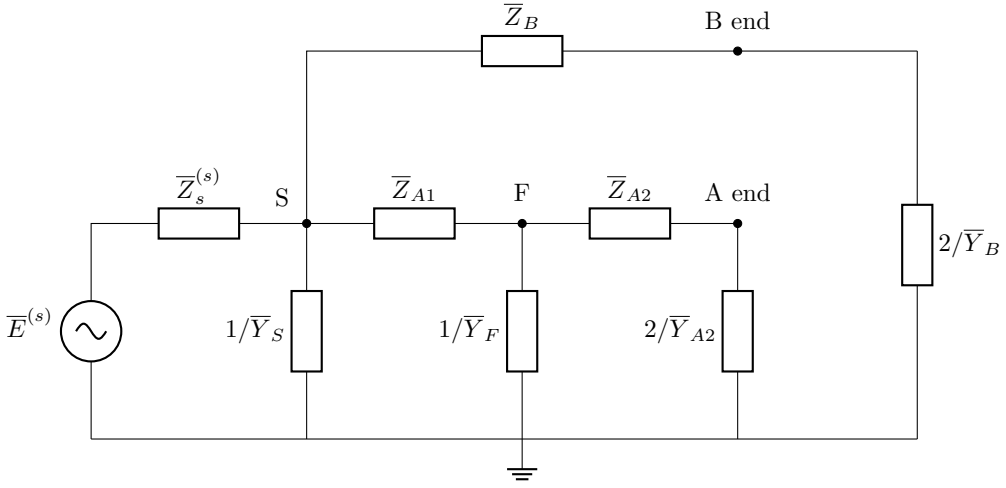


Figure 4. All Π shunts retained, with adjacent half-shunts combined: $\overline{Y}_S = (\overline{Y}_{A1} + \overline{Y}_B)/2$ and $\overline{Y}_F = (\overline{Y}_{A1} + \overline{Y}_{A2})/2$. For $s = 1$, $\overline{E}^{(1)} = \overline{E}$; for $s = 0, 2$, replace the source by a short circuit.

$$\overline{Z}_s^{(1,2)} = \overline{Z}_T \parallel \overline{Z}_T = \overline{Z}_T/2, \tag{25}$$

$$\overline{Z}_s^{(0)} = (\overline{Z}_T + 3\overline{Z}_{\mathrm{ASC}}) \parallel (\overline{Z}_T + 3\overline{Z}_{\mathrm{ASC}}). \tag{26}$$

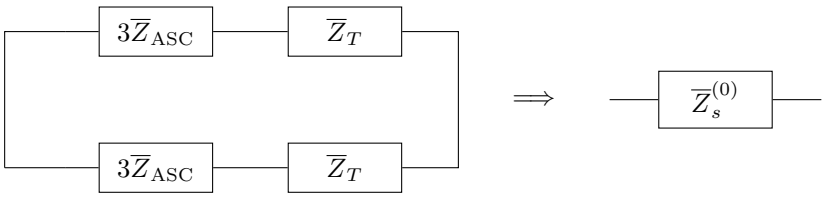


Figure 5. First reduction: the two neutral/transformer branches in parallel.

4.2 Reduce the unloaded terminal sections

For the unloaded sections B and A2:

$$\overline{Y}_{B,\mathrm{in}} = \frac{\overline{Y}_B}{2} + \left(\overline{Z}_B + \frac{2}{\overline{Y}_B}\right)^{-1} = (0.006397 + \mathrm{j}0.597012)\,\mathrm{mS}, \tag{27}$$

$$\overline{Y}_D = \frac{\overline{Y}_{A2}}{2} + \left(\overline{Z}_{A2} + \frac{2}{\overline{Y}_{A2}}\right)^{-1} = (0.009072 + \mathrm{j}1.760307)\,\mathrm{mS}. \tag{28}$$

D denotes section A2 beyond the fault.

5. Reduction to the fault point and precise tuning

5.1 Bus and fault-point driving impedances

With the source deactivated, for $s = 0, 1, 2$:

$$\overline{Z}^{(s)}_{S,\mathrm{eq}} = \left[\frac{1}{\overline{Z}^{(s)}_{s}} + \frac{\overline{Y}_{A1}}{2} + \overline{Y}_{B,\mathrm{in}}\right]^{-1}, \tag{29}$$

$$\overline{Z}^{(s)}_{\mathrm{back}} = \overline{Z}_{A1} + \overline{Z}^{(s)}_{S,\mathrm{eq}}, \tag{30}$$

$$\overline{Z}^{(s)}_{F} = \left[\frac{1}{\overline{Z}^{(s)}_{\mathrm{back}}} + \frac{\overline{Y}_{A1}}{2} + \overline{Y}_{D}\right]^{-1}. \tag{31}$$

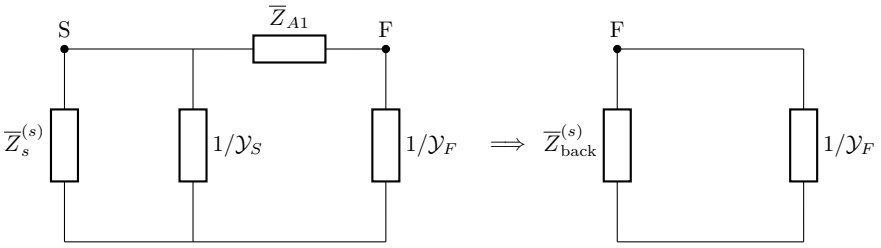


Figure 6. Successive reductions, with $\mathcal{Y}_S = \overline{Y}_{A1}/2 + \overline{Y}_{B,\mathrm{in}}$ and $\mathcal{Y}_F = \overline{Y}_{A1}/2 + \overline{Y}_{D}$. The final parallel combination is $\overline{Z}^{(s)}_{F}$.

5.2 Refine the nominal ASC setting for the full Π model

For full compensation in the Π model, tune both coils using:

$$\mathrm{Im}\left\{\frac{1}{\overline{Z}^{(0)}_{F}(X_L)}\right\} = 0. \tag{32}$$

$$\overline{Z}_{\mathrm{ASC}}(X_L) = \left(R^{-1}_{\mathrm{ASC}} + (\mathrm{j}X_L)^{-1}\right)^{-1}. \tag{33}$$

The resulting setting is

$$X_L = 132.1793\,\Omega, \tag{34}$$

$$L = 0.420740\,\mathrm{H}, \tag{35}$$

$$Q_{\mathrm{set}} = \frac{U_n^2}{3X_L} = 1.220564\,\mathrm{MVAr} < 1.700\,\mathrm{MVAr}, \tag{36}$$

$$\overline{Z}_{\mathrm{ASC}} = (2.760281 + \mathrm{j}132.121619)\,\Omega. \tag{37}$$

$R_{\mathrm{ASC}} = 6326.7974\,\Omega$ remains unchanged.

Reduction [Ω]	Positive/negative sequence	Zero sequence (tuned)
$\overline{Z}^{(s)}_{s}$	$0.029494 + \mathrm{j}1.088601$	$4.169916 + \mathrm{j}199.271029$
$\overline{Z}^{(s)}_{S,\mathrm{eq}}$	$0.029632 + \mathrm{j}1.090874$	$12.229264 + \mathrm{j}322.156605$
$\overline{Z}^{(s)}_{\mathrm{back}}$	$1.310632 + \mathrm{j}3.070077$	$13.510264 + \mathrm{j}324.135808$
$\overline{Z}^{(s)}_{F}$	$1.335870 + \mathrm{j}3.093812$	$6957.013027 + \mathrm{j}0$

6. Earth-fault current at 21 km on feeder A

6.1 Positive-sequence prefault voltage at the fault point

Including cable charging:

$$\overline{Y}_{A,\mathrm{in}} = \frac{\overline{Y}_{A1}}{2} + \left[\overline{Z}_{A1} + \left(\frac{\overline{Y}_{A1}}{2} + \overline{Y}_D\right)^{-1}\right]^{-1}, \tag{38}$$

$$\overline{U}_{S,\mathrm{pre}} = \frac{\overline{E}}{1 + \overline{Z}_s^{(1)}(\overline{Y}_{A,\mathrm{in}} + \overline{Y}_{B,\mathrm{in}})} = (12771.413 - \mathrm{j}2.466)\,\mathrm{V}, \tag{39}$$

$$\overline{E}_F = \frac{\overline{U}_{S,\mathrm{pre}}}{1 + \overline{Z}_{A1}(\overline{Y}_{A1}/2 + \overline{Y}_D)} = (12849.269 - \mathrm{j}53.877)\,\mathrm{V}. \tag{40}$$

6.2 Connect the sequence equivalents in series

For a phase-a earth fault, $\overline{I}_F^{(1)} = \overline{I}_F^{(2)} = \overline{I}_F^{(0)}$ and $\overline{I}_f = 3\overline{I}_F^{(0)}$. With $R_f = 0$,

$$\overline{I}_F^{(0)} = \frac{\overline{E}_F}{\overline{Z}_F^{(1)} + \overline{Z}_F^{(2)} + \overline{Z}_F^{(0)} + 3R_f}. \tag{41}$$

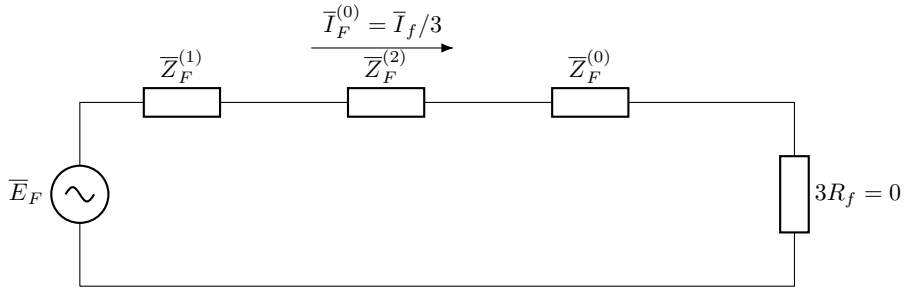


Figure 7. Final series connection of the three sequence equivalents for a single-phase earth fault.

$$\overline{Z}_{\mathrm{loop}} = (6959.684767 + \mathrm{j}6.187624)\,\Omega, \tag{42}$$

$$\overline{I}_F^{(0)} = (1.846235 - \mathrm{j}0.009383)\,\mathrm{A}, \tag{43}$$

$$\overline{I}_f = (5.538704 - \mathrm{j}0.028148)\,\mathrm{A} = 5.538775\angle(-0.291182^\circ)\,\mathrm{A}. \tag{44}$$

6.3 Comparison with the lecture's lumped approximation

If series impedances and charging-voltage differences are neglected,

$$G_\Sigma = g'(l_A + l_B) = 27.000\,\mu\mathrm{S}, \tag{45}$$

$$I_{f,\mathrm{lumped}} = E\left(3G_\Sigma + \frac{2}{R_{\mathrm{ASC}}}\right) = 5.044047\,\mathrm{A}. \tag{46}$$

The final result uses the complete Π model.

At the nominal (unrefined) setting, $|I_f| = 5.820985\,\mathrm{A}$.

7. Zero-sequence quantities at the feeder relays

7.1 Voltage recovery from F to the common bus

The zero-sequence voltage at F is

$$\overline{U}_F^{(0)} = -\overline{Z}_F^{(0)}\overline{I}_F^{(0)} = (-12844.278 + \mathrm{j}65.276)\,\mathrm{V}. \tag{47}$$

Recover the bus voltage through A1 using the bus equivalent:

$$\overline{U}_S^{(0)} = \overline{U}_F^{(0)}\frac{\overline{Z}_{S,\mathrm{eq}}^{(0)}}{\overline{Z}_{A1} + \overline{Z}_{S,\mathrm{eq}}^{(0)}} = (-12764.114 + \mathrm{j}17.458)\,\mathrm{V}, \tag{48}$$

$$\overline{U}_{0,A} = \overline{U}_{0,B} = 12.764126\angle 179.921636^\circ\,\mathrm{kV}. \tag{49}$$

Both relays share the bus voltage. The primary residual voltage is

$$3\overline{U}_0 = (-38.292343 + \mathrm{j}0.052373)\,\mathrm{kV}.$$

All relay values are primary quantities.

7.2 Outgoing feeder currents

Current is positive from the bus into the feeder. Include the near-end half-shunt:

$$\overline{I}_{0,A} = \frac{\overline{U}_S^{(0)} - \overline{U}_F^{(0)}}{\overline{Z}_{A1}} + \frac{\overline{Y}_{A1}}{2}\overline{U}_S^{(0)} = (1.344306 - \mathrm{j}56.407622)\,\mathrm{A}, \tag{50}$$

$$\overline{I}_{0,B} = \overline{Y}_{B,\mathrm{in}}\overline{U}_S^{(0)} = (-0.092077 - \mathrm{j}7.620214)\,\mathrm{A}. \tag{51}$$

Primary quantity	Magnitude	Angle relative to $\overline{E}$
$\overline{I}_{0,A}$	56.423639 A	-88.634786°
$3\overline{I}_{0,A}$	169.270916 A	-88.634786°
$\overline{I}_{0,B}$	7.620770 A	-90.692288°
$3\overline{I}_{0,B}$	22.862311 A	-90.692288°

The residual current is $\overline{I}_a + \overline{I}_b + \overline{I}_c = 3\overline{I}_0$.

7.3 Consistency checks

For current from S towards the neutral/transformer branches,

$$\overline{I}_{0,N} = \frac{\overline{U}_S^{(0)}}{\overline{Z}_s^{(0)}} = (-1.252229 + \mathrm{j}64.027836)\,\mathrm{A}, \tag{52}$$

$$\overline{I}_{0,A} + \overline{I}_{0,B} + \overline{I}_{0,N} = 0. \tag{53}$$

The phase-a fault voltage also satisfies

$$\overline{U}_{a,F} = \overline{E}_F - \left(\overline{Z}_F^{(1)} + \overline{Z}_F^{(2)} + \overline{Z}_F^{(0)}\right)\overline{I}_F^{(0)} = 0. \tag{54}$$

8. Conclusion

Result	Value
ASC rated power (each)	1.700 MVAr
ASC active loss at reference voltage (each)	25.500 kW
ASC parallel loss resistance (each)	6.326797 kΩ
Nominal lumped tuning (each)	$X_L = 133.4633\,\Omega$, $L = 0.424827\,\mathrm{H}$
Refined Π-model tuning (each)	$X_L = 132.1793\,\Omega$, $L = 0.420740\,\mathrm{H}$
Refined setting at $U_n/\sqrt{3}$ (each)	$Q_{\mathrm{set}} = 1.220564\,\mathrm{MVAr}$
Fault current, compensated state	$5.538775\angle(-0.291182°)$ A
Relay zero-sequence voltage, A and B	$12.764126\angle 179.921636°$ kV
Relay residual voltage, A and B	$38.292379\angle 179.921636°$ kV
Feeder A zero-sequence current	$56.423639\angle(-88.634786°)$ A
Feeder A residual current	$169.270916\angle(-88.634786°)$ A
Feeder B zero-sequence current	$7.620770\angle(-90.692288°)$ A
Feeder B residual current	$22.862311\angle(-90.692288°)$ A

9. Phasor diagrams

Voltage phasors: $\arg \overline{E} = 0°$.

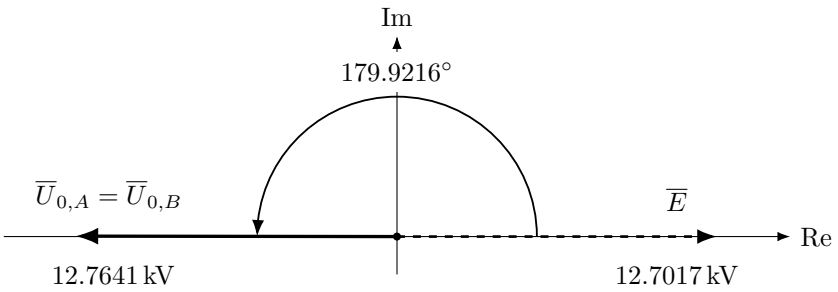
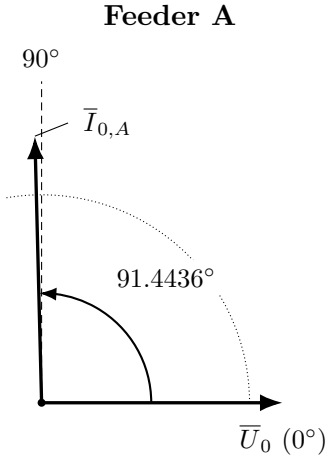


Figure 8. Voltage phasors

Relay phasors: angles measured counterclockwise from $\overline{U}_0$.



$|U_0| = 12.7641\,\mathrm{kV}$
$|I_{0,A}| = 56.4236\,\mathrm{A}$

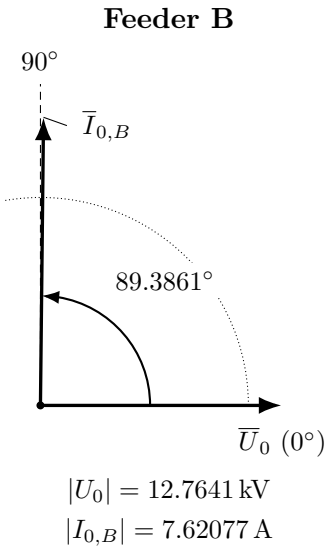


Figure 9. Relay voltage and current phasors at their true phase angles.

$$\varphi_A = \arg\left(\frac{\overline{I}_{0,A}}{\overline{U}_0}\right) = 91.4436^\circ,$$
$$\varphi_B = \arg\left(\frac{\overline{I}_{0,B}}{\overline{U}_0}\right) = 89.3861^\circ.$$

Voltage and current have separate magnitude scales. The residual quantities $3\overline{U}_0$ and $3\overline{I}_0$ have the same angles.